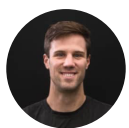


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A Guide to Rep Metrics Available in Metric

Metric tracks 12 different rep-level data points, providing insight into every aspect of barbell motion on every rep. This guide covers each metric



Jacob Tober
Co-founder

Metric tracks a number of different rep-level data points, providing comprehensive insight into every aspect of barbell movement. This guide covers each metric, explaining what it measures, when to use it, and what values to expect.

All metrics are tracked based on the position and speed of the barbell and its weight plates.

A Quick Guide to This Documentation

This blog post focuses on raw rep-level data points, all of the metrics covered in this post are directly calculated from how metric tracks the barbell in space during each repetition. Future posts will explore set-level data points (such as velocity loss) and composite workout scores (such as estimated 1RM) which are calculated from these rep metrics to provide deeper insights into performance as a whole.

For each metric we will cover a short definition and explanation for how it is calculated, provide a recommended use case, typical ranges and then two interpretation guides.

Absolute interpretation (what it means)

Explains how to understand the scores you are seeing in a pure sense, comparing a set with universal standards or between sets with different weights or two different athletes.

Contextual interpretation (how to use it)

Explains how your performance has changed over time, and what those changes mean for fatigue, adaptation and ultimately progress in training. Looking at data points contextually (comparing your best rep performance today with your 6-week average or all-time personal best for the same exercise and weight) is arguably the most valuable use of barbell tracking data — providing real time insights into performance, fatigue, adaptation and program effectiveness.

TL;DR – Rep Metrics Overview

Metric offers 13 rep-level tracking metrics. It is recommended that for each exercise you focus on only 1-2 (3 at the absolute maximum) metrics and ignore all others. For this reason you can only select 3 metrics at a time in the set review screen.

Note: Metric also collects set level and workout level metrics such as estimated 1RM, maximum power output, total time under tension and tonnage scores, we will write a dedicated article covering these in the near future.

Velocity Metrics

1. **Mean Velocity** – *Primary metric recommended for all training* – Average speed throughout entire lift (best for heavy training 80%+ 1RM)
2. **Peak Velocity** – Maximum instantaneous speed (ideal for explosive/Olympic lifting)
3. **Time to Peak** – How quickly maximum velocity is reached (explosive capacity and rate of force development (RFD) indicator)

Power Metrics

4. **Mean Power** – *Default power metric* – Average power throughout entire movement
5. **Peak Power** – Maximum instantaneous power output
6. **Eccentric Power** – Energy absorption during braking

Movement Analysis

7. **Range of Motion** – Total vertical barbell displacement (technique consistency)
8. **Bar Path** – Not technically a metric but a subjective and visual guide to your lifting technique

Tempo Metrics

9. **Eccentric Tempo** – Duration of lowering phase
10. **Concentric Tempo** – Duration of upward phase
11. **Bottom Pause** – Rest time at bottom of a rep (may also include rest time when barbell is on the ground)
12. **Top Pause** – Rest time at bottom/top positions
13. **Time Under Tension** – Total rep duration from start to finish

For simple, focused training: Focus on mean velocity/power for tracking force production and range of motion/eccentric tempo for control and technical consistency. That's all you really need!

Concentric Velocity Metrics

Mean Velocity

Recommended primary velocity metric for all training

This tracks your average barbell speed from the bottom to the top of each rep – basically how fast you're moving the weight throughout the

entire lift.

Units: Meters per second (m/s)

Best for: All training loads and exercises, research comparison, general movement speed assessment, autoregulation

What it means: If you're giving maximum effort and trying to move fast, slower mean velocities tell you you're getting closer to failure. The heavier the weight gets, the slower you'll move it – this relationship is so consistent that you can actually use it to predict your 1RM with a load-velocity profile.

How to use it: As you get stronger, you should be able to lift the same weight faster than before. If your mean velocity is higher than usual, you're producing more force and power. If it's lower than your typical performance at that weight, you might be fatigued, not trying hard enough, or having technique issues.

What to expect:

- Mean velocity at 100% of 1RM: 0.2–0.4 m/s
- Mean velocity at 80%: 0.4–0.7 m/s
- Mean velocity at 60%: 0.8–1.2 m/s

Peak Velocity

Your highest speed reached at any single moment during the lift, usually this occurs near the top or the moment just before leaving the ground/the bar floats.

Units: Meters per second (m/s)

Best for: Supplementary explosive training insights, Olympic lifting analysis

What it means: Peak velocity shows your maximum speed capacity on that exercise and load. Higher peak velocities mean you have better explosive qualities and can recruit your fast-twitch muscle fibers more effectively. However, peak velocity is prone to noise and inconsistent readings due to equipment vibrations, bouncing, and other measurement artifacts that make it less reliable for day-to-day training decisions.

How to use it: While peak velocity can provide insights into explosive power development, its inherent variability makes it challenging to track meaningful changes over time. Mean velocity is strongly correlated with peak velocity ($r^2 = 0.94$), providing similar insights with much greater consistency and reliability.

What to expect:

- Peak velocity at 90% 1RM: 0.3–0.6 m/s
- Peak velocity at 70% 1RM: 0.8–1.4 m/s
- Peak velocity at 50% 1RM: 1.2–2.0 m/s

Why We Don't Track Propulsive Velocity

While propulsive velocity (average speed during only the accelerating portion of each rep) is theoretically interesting and appears in some research, we've chosen not to include it in Metric for several practical reasons:

Technical limitations: Propulsive velocity requires reverse-engineering acceleration data from position measurements. Accurately

determining the precise moment when acceleration transitions to deceleration is extraordinarily difficult, regardless of the tracking technology used. This leads to inconsistent and unreliable measurements that can mislead rather than inform training decisions.

Practical redundancy: Our analysis of thousands of training sessions shows that mean velocity provides the same actionable insights as propulsive velocity while being far more stable and reliable. Since mean velocity captures your effort level and training stimulus effectively across all loads and exercises, adding propulsive velocity would create unnecessary complexity without meaningful benefits.

Focus on what works: We prioritize metrics that provide clear, consistent feedback you can trust session after session. Mean velocity delivers this reliability, making it the superior choice for practical training applications.

Time to Peak Velocity

Also called time to peak for short. Measures how quickly you reach your peak velocity moment during each rep.

Units: Seconds (s)

Best for: Rate of force development assessment, Olympic lifting, explosive training for athletic speed and power

What it means: Shorter times to peak mean you have better rate of force development and explosive qualities. This metric shows how well your nervous system can rapidly recruit motor units and generate force. Keep in mind that values depend heavily on the exercise you're doing, your range of motion, and the load you're using.

How to use it: If your time to peak keeps getting shorter (faster), you're

improving your explosive capacity and making neuromuscular adaptations. If it's getting longer (slower), you might be accumulating fatigue or need to add more explosive training to your program. This metric is particularly sensitive to changes in your training status, so it's great for monitoring progress.

What to expect:

- Olympic lifts (70% 1RM): 0.08–0.20s
- Squats (70% 1RM): 0.20–0.45s
- Bench press (70% 1RM): 0.15–0.35s

Movement Analysis Metrics

Range of Motion

Tracks the total vertical distance the barbell travels during the concentric portion of each rep, measured from your lowest point to the highest point of the lift.

Units: Centimeters (cm)

Best for: Technique consistency, depth monitoring, comparing your reps across different training sessions

What it means: Range of motion reflects how deep you're going and how consistently you're executing the movement. Consistent ROM values mean you have stable movement patterns, while big variations might suggest technique inconsistencies, fatigue effects, or mobility limitations. The optimal ROM is specific to each exercise and varies between individuals.

How to use it: If your ROM is increasing over time, you might be improving your mobility or refining your technique. If it's decreasing, this often signals fatigue, poor movement quality, or compensation patterns developing. Sudden ROM changes are a red flag that you should assess your technique and possibly adjust your load. ROM tracking is also really useful in rehabilitation and return-to-play scenarios. Programming with ROM can also be a great tool in rehabilitation, using progressively deeper ROMs an athlete recovers from an injury.

What to expect:

- Back squat: 0.3–0.7m (highly individual based on limb length)
- Bench press: 0.2–0.5m
- Deadlift: 0.3–0.6m
- Sumo Deadlift: 0.2–0.45m

Bar Path

Shows the visual trajectory of your barbell during each rep – essentially drawing a line that tracks exactly how the bar moves through space from start to finish.

Units: Visual trajectory (not a numerical metric)

Best for: Technique analysis, movement consistency, identifying compensation patterns

What it means: Bar path reveals the "shape" of your lift and shows whether you're moving the bar efficiently. An optimal bar path is typically as vertical as possible for most exercises, though this varies

by lift type. Deviations forward, backward, or side-to-side can indicate technique issues, mobility limitations, or compensation patterns.

How to use it: Consistent bar paths between reps indicate stable technique and good movement control. If you notice your bar path changing – drifting forward on squats, pressing forward on bench, or swaying during deadlifts – this often signals technique breakdown, fatigue, or the need for mobility work. Comparing bar paths across different sessions helps you identify whether technique changes are temporary (fatigue-related) or developing patterns that need attention.

What to look for:

- **Squats:** Nearly vertical path with minimal forward drift
- **Bench Press:** Slight backward curve from centre of chest to over shoulders
- **Deadlifts:** Vertical path staying close to your body
- **Overhead Press:** Straight vertical path once the bar clears your head
- **Olympic lifts:** have a distinct curved and looping shape, this is individual specific and tightens with training.

Bar path visualization is particularly valuable because it gives you immediate visual feedback that's easy to understand – you can literally see when a rep "looks wrong" compared to your typical movement pattern.

Power Metrics

Mean Power

Recommended primary power metric for all training

Calculates your average power output throughout the entire lift by multiplying force and mean velocity across range of motion.

Units: Watts (W), or can also be in relative watts (W/kg) *Watts produced ÷ your bodyweight

Best for: All power exercises, strength exercises done with dynamic effort, exercises with accommodating resistance (bands and chains), overall power assessment, autoregulation

What it means: Mean power shows your ability to produce force rapidly throughout the complete movement. Higher values indicate you have better strength-speed characteristics. Since power combines both load and velocity, both components will influence your power score. This provides a comprehensive view of your force production capabilities that's consistent and reliable across all exercises and loads.

How to use it: If your mean power increases at the same load, you're improving your force production and rate of force capacity. Track your mean power across all loads for a given exercise to find where you produce maximum power output. Your maximum mean power typically occurs at moderate loads (40-60% 1RM), and tracking power across different loads helps you identify the ideal training load for power development and expression.

What to expect: These ranges are highly individual and exercise-dependent

- Elite athletes: 2000-4000W+ on lower body maximum power loads

- Intermediate athletes: 1200–3000W at maximum power loads
- Novice athletes: 600–1500W at maximum power loads

Peak Power

Measures your highest instantaneous power output during the lift, usually just before leaving the ground or when the bar reaches a float phase.

Units: Watts (W)

Best for: Supplementary explosive training insights, ballistic exercises analysis

What it means: Peak power shows your maximum power capacity under whatever load you're using. This metric is most meaningful during explosive movements and light-to-moderate loading. However, peak power is susceptible to measurement noise, equipment vibrations, and technique variations that can create inconsistent readings, making it less reliable for consistent training feedback.

How to use it: While peak power can provide insights into explosive capacity, its inherent variability makes it challenging to track meaningful changes over time. Mean power provides similar insights into your power development with much greater consistency and reliability for making training decisions.

What to expect:

- Peak power often occurs at 30–60% 1RM
- Elite athletes may achieve 5000W+ peak power values on certain exercises

- Load for peak power varies significantly between exercises and individuals

Why We Don't Track Propulsive Power

Just like with propulsive velocity Metric does not provide propulsive power metrics. Mean power is more reliable (good for you) easier to measure (good for the app) and will do more than well enough for all training contexts. If you want to start going deeper into the technical aspects of your power training then peak metrics or time to peak velocity offer valuable insights.

Eccentric Power

Measures how much energy you absorb during the braking phase when controlling a load – calculated using the weight, how fast you're peak eccentric velocity is, and how quickly you can bring the bar to a stop.

Units: Watts (W)

Best for: Drop catch exercises, step downs, eccentric lunge movements, eccentric strength assessment, deceleration training

What it means: Eccentric power shows your ability to rapidly decelerate and control loads. Higher values indicate you have better braking capacity and eccentric strength. This metric only captures power applied by the athlete so when looking at eccentric power avoid letting the barbell be stopped by the ground or by spotting arms.

How to use it: Improvements in eccentric power suggest you're developing better deceleration capacity and eccentric strength. This is

particularly valuable if you play sports that require rapid deceleration, landing, change of direction, and catching movements.

What to expect:

- Regular exercises (no eccentric intent): 10–100W
- Specific eccentric "drop" exercises
 - Novice athletes: 200–300W
 - Intermediate athletes: 300–600W
 - Advanced athletes: 600W+

Tempo Metrics

Eccentric Tempo

Tracks the duration of your lowering phase for each rep

Units: Seconds (s)

Best for: Time under tension control, technique consistency, strength endurance, hypertrophy and volume accumulation phases of training

What it means: Longer eccentric tempos increase muscle damage and metabolic stress, while shorter tempos emphasize power and explosiveness. Consistent tempos indicate good technique control.

What to expect:

- Normal tempo training: 0–2s

- Controlled tempo training: 2–5s
- Slow eccentric training: 4–8s

Concentric Tempo

Measures how long your lifting phase takes, from the bottom of the movement until you complete the upward phase.

Units: Seconds (s)

Best for: Grinding assessment, strength endurance, hypertrophy and volume accumulation phases of training

What it means: Longer concentric tempos increase muscle damage and metabolic stress, and contribute to your total time under tension for a muscle group when training for hypertrophy. Consistent tempos indicate good technique control.

What to expect:

- Light loads (50–60% 1RM): 0.4–1.0s
- Moderate loads (70–80% 1RM): 0.8–2.0s
- Heavy loads (85–95% 1RM): 1.5–4.0s

Bottom Pause

Tracks how long you spend stationary at the bottom position between the lowering and lifting phases.

Units: Seconds (s)

Best for: Pause technique validation, eliminating stretch–shortening cycle

What it means: Longer pauses eliminate more elastic energy and make the subsequent lift harder. Consistent pause durations ensure you're getting a standardized training stimulus and not cheating the movement. Bottom pause durations will also include time when the barbell is resting on the ground for deadlift and row variations.

What to expect:

- Competition pause requirements: 1-2s
- Training pause protocols: 2-5s

Top Pause

Measures how long you rest at the top position before starting the next rep or finishing the set.

Units: Seconds (s)

Best for: Rest-pause training, technique analysis, set completion verification

What it means: Longer pauses between reps often indicate accumulating fatigue. This metric helps identify when you need additional rest between reps and can verify accurate rep counting.

What to expect:

- Normal training: 0.5-2.0s between reps
- Rest-pause protocols: 10-20s
- Heavy strength training: 2-8s between reps

Time Under Tension

Tracks the total time from when you start lowering the weight until you complete the lift – the complete time your muscles are working during each rep.

Units: Seconds (s)

Best for: Hypertrophy training, metabolic stress assessment, overall rep duration

What it means: Longer time under tension increases metabolic stress and may enhance muscle growth. Optimal ranges depend on your training goals – power training uses shorter durations while hypertrophy training uses longer ones.

What to expect:

- Power training: 1–3s
- Strength training: 2–6s
- Hypertrophy training: 4–8s
- Endurance training: 6–12s

Using Rep Metrics Effectively

Focus on What Matters Most

Remember: you don't need to track all metrics at once. Pick 1–2 metrics for each exercise and focus on those – all metrics are being tracked in the background for all sets with the Metric app, so if you decide to switch later, your historical data and context will be readily available. This focused approach lets you actually use the data to improve your

training instead of getting overwhelmed by numbers.

Essential Metrics for Most Lifters

I strongly recommend using only one form of concentric velocity metric across all your lifts: **mean velocity**.

It provides reliable, consistent feedback across all exercises and loads while avoiding the measurement complexities and noise issues that plague other velocity metrics. Plus using only one metric at all times cuts 100% of the cognitive load associated with juggling multiple velocity metrics.

Yes peak velocity is technically superior for explosive lifts, but the added complexity of switching metrics for different exercises and the notorious instability of peak metrics far outweighs the benefits!

If you already have a good practice for using peaks in your training then carry on, they remain available in the Metric app and are still valuable metrics. But if you're just starting out or looking to simplify your approach, here's what I recommend focusing on:

- **Mean velocity** (your primary velocity metric for everything)
- **Range of motion** (technique consistency and depth monitoring)
- **Mean power** (power development and load optimization)

These three metrics will give you everything you need for 90% of your training decisions. I use these metrics almost exclusively in my training.

The beauty of this approach is its simplicity and reliability. Mean velocity works consistently whether you're doing heavy singles at 95%

1RM or explosive training at 60% 1RM. You never have to switch metrics or remember which one works best for which exercise – just focus on lifting and let the data guide your decisions.

Adding Specialization

Once you're comfortable with the essentials, you can incorporate additional metrics based on your specific training focus. You might use these on an exercise-specific basis or rotate them in and out of your training plan during different training phases or blocks depending on the demands of your program.

It can also just be good fun to rotate metrics and chase a different quality for a few weeks, giving the same exercises new meaning!

Explosive Training: Add time to peak velocity to monitor rate of force development

Eccentric Focus: Add eccentric power for deceleration and braking capacity

Technique Work: Add bottom pause and eccentric tempo for movement quality

Heavy Strength Training: Monitor bottom pauses, range of motion, and concentric tempo when you're truly grinding through reps close to the point of failure.

Making Meaningful Comparisons

The real utility of barbell tracking comes from comparing your current performance to your own training history. Here's how to do it right:

Track your personal patterns

Look at your recent averages, personal bests, and trends over the past few weeks or months. Tracking personal records is a great performance indicator for velocity and power, while [the Velocity Trend Score](#) is a great way to track your fatigue and adaptation over the last 6-weeks

Compare like with like

Always compare metrics at similar loads and intensities. Your velocity at a 100kg back squat should be compared to your historical velocity at 100kg, not 110kg and not to a front squat. Velocity is particularly exercise and load specific.

Consider your context

Even your own recent historical data can be misleading. Remember to factor in your current training state and fatigue levels as well.

Depending on where you are in your training cycle a "slow" velocity might actually be impressive if you're in a high-volume phase or coming back from a break.

This contextual approach – seeing how today's performance stacks up against your own patterns – provides the most valuable insights for optimizing your training outcomes and making real progress in the gym.

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By lifters for lifters

Metric was built with lifters and coaches front of mind.

Focused analytics and powerful workout tracking tools for Powerlifting, Strength and conditioning, CrossFit, Weightlifting, and anyone serious about their strength training.



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